

8 DSA Patterns

jo 80% coding interviews crack kara dete hain

01 Two Pointers

Sorted array · pair dhoondhna · reverse · remove duplicates · in-place

KEYWORDS

sorted pair sum closest remove duplicates
palindrome reverse

INTUITION

Do pointers different positions se start hote hain aur unnecessary comparisons bachate hain.

TIME

$O(n)$

FAMOUS PROBLEMS

- Two Sum II
- Valid Palindrome
- Container With Most Water
- Remove Duplicates from Sorted Array
- Move Zeroes

02 Sliding Window

Continuous subarray / substring problems

KEYWORDS

longest shortest maximum minimum
consecutive contiguous window at most k
exactly k

INTUITION

Ek window maintain karo, condition toot jaaye to left pointer badhao.

TIME

$O(n)$

FAMOUS PROBLEMS

- Maximum Sum Subarray
- Longest Substring Without Repeating Characters
- Minimum Window Substring
- Fruits Into Baskets

03 Fast & Slow Pointer

Linked list ya cycle detection

KEYWORDS

cycle middle node happy number circular
detect loop

INTUITION

Ek pointer slow, ek fast — agar mil gaye, cycle hai.

TIME

$O(n)$

FAMOUS PROBLEMS

- Linked List Cycle
- Middle of Linked List
- Happy Number
- Circular Array Loop

04 Binary Search

Sorted search space ya monotonic answer

KEYWORDS

sorted minimum possible maximum possible
search peak rotated first/last occurrence

INTUITION

Har step pe aadha search space hata do.

TIME

$O(\log n)$

FAMOUS PROBLEMS

- Search Insert Position
- Koko Eating Bananas
- Capacity To Ship Packages
- Search in Rotated Sorted Array

05 Prefix Sum

Multiple range sum queries

KEYWORDS

range sum sum between L and R count subarrays
equal sum prefix

INTUITION

Current sum = Previous prefix + current element.

PREPROCESSQUERY

$O(n)$ $O(1)$

06 DFS / BFS

Tree, graph, matrix traversal

KEYWORDS

connected island shortest path traversal
visit flood fill component

INTUITION

DFS → deep pehle jao. BFS → level by level jao.

TIME

$O(V+E)$

FAMOUS PROBLEMS

- Range Sum Query
- Subarray Sum Equals K
- Find Pivot Index

FAMOUS PROBLEMS

- Number of Islands
- Flood Fill
- Binary Tree Level Order
- Clone Graph

Backtracking

Saari possibilities generate karni ho

KEYWORDS

all combinations all subsets permutations
sudoku n queens generate

INTUITION

Choose → explore → undo → repeat.

TIME

mostly exponential

FAMOUS PROBLEMS

- Subsets
- Combination Sum
- Permutations
- N Queens
- Sudoku Solver

Dynamic Programming

Repeated subproblems + optimal answer chahiye

KEYWORDS

maximum minimum count ways best rob climb
longest

INTUITION

Ek hi problem baar baar solve mat karo, store kar lo.

TIME

mostly $O(n)$ / $O(n^2)$

FAMOUS PROBLEMS

- Climbing Stairs
- House Robber
- Coin Change
- Longest Increasing Subsequence
- 0/1 Knapsack

🔥 Reel cheat sheet — quick recall table

PATTERN	REEL KEYWORDS
Two Pointers	sorted, pair, remove, reverse
Sliding Window	longest, shortest, subarray, substring
Fast & Slow	cycle, middle, loop
Binary Search	sorted, search, minimum, maximum
Prefix Sum	range sum, prefix, subarray sum
DFS / BFS	island, tree, graph, matrix
Backtracking	all combinations, generate, sudoku
Dynamic Programming	maximum, minimum, count ways, longest